

Task Manager (Python CLI)

A small, dependency-free command-line task manager. Create, prioritise, tag, and track tasks from your terminal, with tasks persisted to a local JSON file.

Features

- Create tasks with a title, description, priority (1-4), due date, and tags
- List tasks, with filters by status, priority, or "overdue only"
- Update status (`todo`, `in_progress`, `review`, `done`), priority, and due date
- Add and remove tags
- Show full details of a single task
- Delete tasks
- View statistics (counts by status/priority, overdue, completed in the last 7 days)

Requirements

- Python 3.8+
- No third-party dependencies (standard library only)

Installation

```
git clone <repo-url>
cd TaskManager
# no dependencies to install
python cli.py --help
```

Tasks are stored in `tasks.json` in the working directory (created on first write).

Usage

```
# Create a task (priority 3 = high, with a due date and tags)
python cli.py create "Write report" -d "Q3 summary" -p 3 -u 2026-08-15 -t "work,urgent"

# List tasks
python cli.py list
python cli.py list -s todo           # only TODO
python cli.py list -p 4             # only URGENT
python cli.py list -o               # only overdue

# Update a task (use the id shown by 'list'/'create')
python cli.py status <task_id> done
python cli.py priority <task_id> 4
python cli.py due <task_id> 2026-09-01
```

```
# Tags
python cli.py tag <task_id> blocker
python cli.py untag <task_id> blocker

# Inspect and remove
python cli.py show <task_id>
python cli.py delete <task_id>

# Statistics
python cli.py stats
```

Priority values: 1 = LOW, 2 = MEDIUM (default), 3 = HIGH, 4 = URGENT. **Date format:** YYYY-MM-DD .

Project structure

```
TaskManager/
├─ cli.py           # argparse command-line interface (entry point)
├─ task_manager.py  # service layer: CRUD, filtering, statistics
├─ storage.py       # persistence: JSON load/save, enum & datetime (de)serialisation
├─ models.py        # domain: Task, TaskPriority, TaskStatus
├─ task_priority.py # importance scoring / ranking
├─ task_parser.py   # natural-language task parsing (@tag !priority #date)
└─ task_list_merge.py # two-way sync / conflict resolution
```

Configuration

Persistence path defaults to `tasks.json` . It is set where `TaskManager / TaskStorage` are constructed (`storage_path` argument).

Troubleshooting

- **"Invalid date format. Use YYYY-MM-DD"** — dates must be exactly `YYYY-MM-DD` .
- **"Task not found"** — pass the id shown by `list/create` (the display truncates to 8 chars, but commands expect the full id).
- **Changes not saved** — ensure the process has write permission in the working directory for `tasks.json` .

License

MIT (placeholder — confirm with the project owner).